

Member ID: _____

Time: _____

Rank: _____



COMPUTER SECURITY (320)

NATIONAL 2023

Total Points (50 @ 2 points each) _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. Which of the following would be a typical component of a well-defined Service Level Agreement?
 - A. Type of service to be provided
 - B. Time frame of issue resolution
 - C. Penalties for not meeting the requirements of the SLA
 - D. All of the others answers are correct
2. _____ is a set of privacy and security rules meant to protect patient information relating to health.
 - A. HIPAA
 - B. CDC
 - C. OSHA
 - D. None of the other answers are correct
3. Which of the following descriptions is accurate for the network diagnostic tool, tracert?
 - A. Provides a list of IP addresses and port numbers
 - B. Sends an ICMP packet to a single device
 - C. Sends an ICMP packet across multiple devices and counts hops
 - D. Outputs a network diagram using LLDP
4. When a company establishes security policies, what is the *most* effective way to improve security?
 - A. Disable the Internet access
 - B. Require two-factor authentication
 - C. Provide employee-based user-awareness training
 - D. Enforce account lockout
5. Destruction of data drives is crucial to any company's security policy. Which of the following methods should be used to *best* prevent data extraction from the disks?
 - A. Degauss the drives
 - B. Delete the data on the disks
 - C. Shred the drives
 - D. Write junk data over the disk several times
6. You have hired 20 temporary employees for a three-month period. What should the network administrator do to ensure a high level of security when their employment is terminated after three months?
 - A. Configure account expiration in the user accounts
 - B. Configure account policies in Group Policy
 - C. Configure day / time restrictions in the user accounts
 - D. Configure account lockout in Group Policy

7. You have been tasked with making the network more manageable; in the process you see that the research department is on the same network segment as the human resources department. What can be done to isolate these departments?
 - A. Implement the principle of least privilege for the HR department
 - B. Identify the choke points in your network
 - C. Move the sales department into the DMZ
 - D. Create a separate VLAN for each department
8. An employee loses their company-provided smart phone at the airport, and the device *cannot* be recovered, which of the following will ensure data confidentiality on the device?
 - A. TPM
 - B. CPS
 - C. Screen lock
 - D. Remote wipe
9. What is the primary benefit of CCTV?
 - A. Increases security protection throughout an environment
 - B. Expands the area of visibility of security guards
 - C. Reduces the needs for locks and sensors on doors
 - D. Allows companies to reduce their security staff
10. Symmetric encryption is faster than asymmetric encryption.
 - A. True
 - B. False
11. What type of attack is *most likely* to succeed with communications between instant messaging clients?
 - A. Sniffing
 - B. DNS poisoning
 - C. Brute force password attack
 - D. DOS
12. Which of the following attacks, if successful, causes a switch to function like a hub?
 - A. MAD spoofing
 - B. MAC flooding
 - C. Relay
 - D. ARP poisoning
13. _____ is a method in which an attacker sends invalid ARP messages into a LAN, with the goal of associating their MAC address with the IP address of another host?
 - A. ARP spoofing
 - B. ARP cache poisoning
 - C. ARP poison routing
 - D. All of the answers are correct

14. Which of the following is the primary difference between encryption and hashing?
- A. Encryption is used to convert readable data into an unreadable format
 - B. Hashing is used to convert readable data into an unreadable format
 - C. Encrypted data cannot be converted back to the original data
 - D. Hashed data cannot be converted back to the original data
15. Which of the following router management options would be considered the most secure?
- A. HTTP
 - B. Telnet
 - C. SSH
 - D. None of the other answers are correct
16. Which of the following devices can monitor a network and detect intrusions, but is unable to proactively prevent them?
- A. IDS
 - B. L2 switch
 - C. IPS
 - D. SIEM
17. _____ is an application that is mean to display which ports are open on a network or server.
- A. Port scanner
 - B. Network diagram
 - C. Packet tracer
 - D. None of the other answers are correct
18. Which of the following is *not* a valid Windows password complexity rule for Microsoft accounts?
- A. Password must be eight or more characters long
 - B. Password must contain uppercase characters
 - C. Password must contain spaces
 - D. Password must contain special characters
19. TACACS+ is typically used for centralized remote access and administration. Which protocol is used to accomplish this task?
- A. UDP
 - B. SNMP
 - C. TCP
 - D. FTP
20. Which of the following ranges is valid for a Class A IP address?
- A. 0.0.0.0 to 127.255.255.255
 - B. 128.0.0.0 to 191.255.255.255
 - C. 192.0.0.0 to 223.255.255.255
 - D. 224.0.0.0 to 239.255.255.255

21. Which of the following is the default subnet mask for a Class B IP Address?
- A. 255.255.255.0
 - B. 255.255.240.0
 - C. 255.255.0.0
 - D. 255.1.0.0
22. Which of the following number of hosts per network is valid for a Class C network?
- A. 126
 - B. 254
 - C. 65,534
 - D. 16,777,214
23. Which protocol is used for accessing directory services?
- A. IPC
 - B. LDAP
 - C. SNMP
 - D. CDP
24. The below output is used to retrieve a list of network connections. What command is used here?

```
[devicename | *] [\\computername\sharename[\volume] [password | *]]  
  [/USER:[domainname\]username]  
  [/USER:[dotted domain name\]username]  
  [/USER:[username@dotted domain name]  
  [/SMARTCARD]  
  [/SAVECRED]  
  [[/DELETE] | [/PERSISTENT:{YES | NO}]]
```

- A. tracert
 - B. get net
 - C. netstat
 - D. net use
25. Which of the following shows the IP address 192.168.1.254 converted to binary?
- A. 11000000.10101000.00000001.11111110
 - B. 11111110.00000001.10101000.11000000
 - C. 3232236030
 - D. 0306322323
26. A ROT cipher simply replaces a letter with the 13th letter after it in the alphabet. Hypothetically, if we were to use a ROT5 cipher on “BPA”, which of the following would be the output?
- A. OCN
 - B. GUF
 - C. LZK
 - D. ESD

27. A _____ is a file that is used to authenticate a device or user, using cryptography and a public key.
- A. hash value
 - B. private key
 - C. digital signature
 - D. digital certificate
28. Which of the following information is contained within the /etc/shadow file?
- A. Username
 - B. Encrypted password
 - C. Last password change
 - D. All of the other answers are correct
29. Which of the following properly describes the steps to create an encrypted message that contains a digital signature using PKI? Place the steps in the correct order.
- 1. Encrypt the message with the recipient's public key
 - 2. Create a hash of the message
 - 3. Create the message to be sent
 - 4. Encrypt the hash with your private key
- A. 1, 2, 3, 4
 - B. 3, 2, 1, 4
 - C. 1, 3, 2, 4
 - D. 3, 2, 4, 1
30. Payment Card Industry Data Security Standard (PCI-DSS) has 12 requirements, which of the following is *not* a valid requirement?
- A. Install and maintain a firewall configuration to protect cardholder data
 - B. Ensure that all transferred data is in plaintext
 - C. Do not use vendor-supplied defaults for system passwords and other security parameters
 - D. Protect stored cardholder data
31. How many bits does an IPv6 address use?
- A. 16 bits
 - B. 32 bits
 - C. 64 bits
 - D. 128 bits
32. The four steps of the IPv6 DHCP process can be abbreviated as which of the following?
- A. SORA
 - B. DOSA
 - C. SARR
 - D. DORA

33. SYN flood attacks occur at the _____ layer of the OSI model?
- A. network
 - B. data link
 - C. physical
 - D. transport
34. You have started penetration testing and are starting to resolve domain names to IP addresses. Which of the following is *correct* syntax to resolve BPA.org to an IP address?
- A. host -t a BPA.org
 - B. host -t mx BPA.org
 - C. host -t ns BPA.org
 - D. host -t soa BPA.org
35. As part of a penetration test, you are scanning the ports of a Linux system. Listed here is the scan you performed: nmap -sX Li -vv -P0 192.168.1.123. If the system had the specific listening port open, what would be returned?
- A. RST
 - B. No response
 - C. SYN ACK
 - D. ACK
36. The _____ file is a Windows file that stores user account information, including hashed passwords.
- A. NTUSER
 - B. LSA
 - C. SAM
 - D. PAGING
37. Which of the following is considered to be a type of biometric system authentication?
- A. Password or pin
 - B. Smart card
 - C. Password in conjunction with a security token
 - D. None of the other answers are correct
38. Which of the following file systems would be considered the most secure for a Windows operating system?
- A. HFS+
 - B. NTFS
 - C. FAT
 - D. FAT32
39. Which of the following devices is used to capture keystrokes that are made on a computer's keyboard?
- A. Botnet
 - B. Phishing
 - C. Keylogger
 - D. Fluke

40. You have been asked to use the Netcat utility to listen for incoming connections on port 25, which of the following is a valid syntax?
- A. nc -z -v localhost 25
 - B. nc -L localhost:25 remotehost:25
 - C. nc -l -p 25
 - D. None of the other answers are correct
41. Which of the following is *not* a common method or tool used for static malware analysis?
- A. PE Viewers
 - B. Debuggers
 - C. Disassemblers
 - D. Notepad++
42. Which of the following is a common encryption algorithm used with symmetric encryption?
- A. MD5
 - B. SHA-256
 - C. AES
 - D. RSA
43. What of the following is the primary difference when comparing a worm and a virus?
- A. A worm requires user interaction to spread, while a virus is automated
 - B. A virus requires user interaction to spread, while a worm is automated
 - C. A worm can infect only Windows systems, while a virus can infect any system
 - D. A virus can infect only Windows systems, while a worm can infect any system
44. Which of the following types of attacks involves repeatedly guessing a password in order to gain unauthorized access to a device?
- A. DoS
 - B. Brute-force
 - C. Phishing
 - D. Man-in-the-middle
45. Which of the following types of attacks is meant to disrupt operations of a network and / or service?
- A. DoS
 - B. Brute-force
 - C. Phishing
 - D. Man-in-the-middle
46. Which of the following is the valid frequency range of the ISM frequency band used by Bluetooth?
- A. 2.400 - 2.4835 GHz
 - B. 6.765 - 6.795 MHz
 - C. 244 - 246 GHz
 - D. 2.4 - 5 GHz

47. Which of the following statements is true when considering the security of different protocols?
- A. Telnet is more secure than SSH
 - B. FTP is more secure than SFTP
 - C. SMBv1 is the most secure version of SMB
 - D. None of the other answers are correct
48. Digging through company waste in an effort to find incriminating evidence or information is called ____.
- A. phishing
 - B. social engineering
 - C. digital forensics
 - D. dumpster diving
49. Which of the following is *not* a valid hashing algorithm?
- A. SHA
 - B. MD5
 - C. BLAKE2
 - D. CryptHash
50. _____ is a Windows feature that is used to perform full-disk encryption.
- A. BitLocker
 - B. TPM
 - C. Dm-crypt
 - D. FileVault